

MACHINE LEARNING PROJECT

Flight Price Prediction

Using Machine Learning ✈️

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Problem Statement



Dynamic Pricing

Flight ticket prices fluctuate frequently based on demand, season, and availability.



Multiple Factors

Prices depend on airline, source/destination, departure time, and days before travel.



Traveler Challenge

Passengers struggle to determine the best time to buy a ticket at a fair price.

 **Goal: Build a Machine Learning model to accurately predict flight ticket prices.**

Dataset Description

~300K

Rows

~10

Columns

Kaggle

Source

2 Classes

Economy / Business

Feature	Description	Type
Airline	Airline company name	Categorical
Source / Destination	Departure & arrival city	Categorical
Departure / Arrival Time	Time of travel	Categorical
Duration	Total flight duration (hrs)	Numerical
Days Left	Days before departure	Numerical
Price	Ticket price (₹) — Target	Numerical

Exploratory Data Analysis (EDA)

Step 01 | Dataset Structure

Verified shape, dtypes, and basic statistics of all 10 columns.

Step 02 | Missing Values

No null values found — dataset was clean and ready to use.

Step 03 | Price Distribution

Right-skewed distribution; majority tickets priced between ₹5K–₹30K.

Step 04 | Days Left vs Price

Prices rise sharply as departure date approaches (< 7 days).

Step 05 | Airline Comparison

Business class is 4-5× more expensive than economy on average.

 *Key Insight: Ticket prices increase significantly as the departure date approaches.*

Data Preprocessing

1

Remove Unnecessary Columns

Dropped irrelevant identifiers that don't contribute to price prediction.

2

Label & One-Hot Encoding

Converted categorical features (airline, city, time) into numerical values.

3

Train-Test Split

80% training data / 20% testing data using scikit-learn's `train_test_split`.

Train / Test Split

Training Data — 80%

Test — 20%

Tools Used:

`pandas` • `scikit-learn` • `numpy`

All preprocessing steps performed using Python libraries

Feature Engineering

Airline

High

Different airlines charge different price premiums

Duration

High

Longer flights generally cost more due to fuel costs

Days Left

Very High

Strongest predictor — last-minute tickets cost much more

Departure Time

Medium

Early morning & late night flights tend to be cheaper

Source & Destination

High

Route popularity greatly affects base pricing

Class

Very High

Business class is 4-5× more expensive than economy

Machine Learning Model

Algorithm Used

Linear Regression

-  Predicts continuous numerical values
-  Simple and highly interpretable
-  Fast training on large datasets
-  Good baseline model for regression

1. Raw Dataset



2. Preprocessing



3. Train Set (80%)

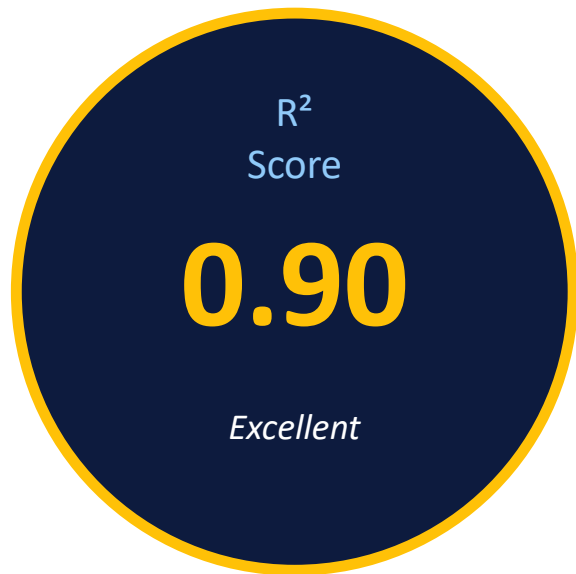


4. Train Model



5. Predictions

Model Performance & Evaluation



What $R^2 = 0.90$ Means

- The model explains 90% of price variation
- Only 10% of variance is unexplained
- Strong predictive capability on unseen data
- Ready for real-world deployment



90% Accuracy

MAE: Measures avg. absolute error

MSE: Penalises large errors heavily

R²: Variance explained by model

Project Implementation

Tech Stack

Python

Core Language

Flask

Web Backend

Gradio

UI Framework

scikit-learn

ML Library

pandas

Data Handling

User Inputs

✓ Flight Duration (hrs)

✓ Days Before Departure

✓ Airline Name

✓ Source & Destination

✓ Travel Class

 **Output**

**Predicted
Ticket Price (₹)**

- Price Prediction Chart
- Airline Price Analysis

Conclusion & Future Work

✓ What We Achieved

- Built a Linear Regression model for predicting ticket prices
- Conducted end-to-end EDA and data preprocessing
- Engineered key features that impact pricing
- Achieved 90% R^2 Score on test data
- Developed interactive UI using Flask & Gradio

🚀 Future Improvements

- Implement advanced models: XGBoost, Random Forest
- Integrate live real-time flight price APIs
- Deploy as full web application on cloud
- Add multi-city & multi-stop route support
- Build personalized alert system for price drops